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UNIT-7

Choose the correct answer:

1. option (c)

$$k = \left(\frac{2.303}{t}\right)\log\left(\frac{[A_0]}{[A]}\right)$$

$$k = \left(\frac{1}{t}\right)\ln\left(\frac{[A_0]}{[A]}\right)$$

$$e^{kt} = \left(\frac{[A_0]}{[A]}\right)$$

$$[A] = [A_0]e^{-kt}$$

In this case

$k = x\,\text{min}^{-1}$ and $[A_0] = 0.01\text{M} = 1 \times 10^{-2}\text{M}$

$t = 1$ hour $= 60$ min

$$[A] = 1 \times 10^{-2}\left(e^{-60x}\right)$$

2. option (c)

for $n \neq 1$ $\quad t_{1/2} = \dfrac{2^{n-1}-1}{(n-1)\,k\,[A_0]^{n-1}}$

for $n = 0$ $\quad t_{1/2} = \dfrac{1}{2\,k\,[A_0]^{-1}}$

$$t_{1/2} = \frac{[A_0]}{2\,k}$$

$t_{1/2} \alpha [A_0]$ ———(1)

given

$[A_0] = 0.02\text{M}$; $t_{1/2} = 10\,\text{min}$

$[A_0] = 0.04\text{M}$; $t_{1/2} = ?$

substitute in (1)

$10\,\text{min} \alpha\, 0.02\text{M}$ ———(2)

$t_{1/2} \alpha\, 0.04\text{M}$ ———(3)

$(3)/(2)$

$$\Rightarrow \frac{t_{1/2}}{10\,\text{min}} = \frac{0.04\ \text{M}}{0.02\ \text{M}}$$

$$t_{1/2} = 2 \times 10\,\text{min} = 20\,\text{min}$$

Answers.indd 243 3/29/2019 6:43:57 PM